Quantification: 699_HYP_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	17
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1788.53	1329	1552.25	px^2
major_axis_length_px	lacuna shape	81.4362	72.5694	43.7243	px
minor_axis_length_px	lacuna shape	33.9902	30.7419	16.8535	px
aspect_ratio	lacuna shape	2.631	2.3763	1.2044	unitless
eccentricity	lacuna shape	0.8616	0.9071	0.1151	unitless
solidity	lacuna shape	0.7568	0.7599	0.1037	unitless
roots_count	headline	6.3529	5	3.6562	count
ring_length_r30_px	headline	222.235	213	143.832	px
ring_length_r60_px	reported	442.353	405	310.73	px
owned_length_px	ownership-dependent	479.245	345.238	478.309	px
edge_count	ownership-dependent	26.1176	17	24.6041	count
mean_edge_length_px	ownership-dependent	17.961	18.924	5.1127	px
perimeter_px	normalised	251.52	221.179	140.119	px
ring_area_r30_px2	normalised	7892.24	6924	3392.08	px^2
ring_area_r60_px2	normalised	17908.9	16630	7007.43	px^2
in_frame_fraction_r30	normalised	0.9773	1	0.0659	unitless
in_frame_fraction_r60	normalised	0.9644	1	0.101	unitless
ring_density_r30	normalised	0.0295586	0.0306743	0.017681	px^-1
ring_density_r60	normalised	0.0243639	0.0282702	0.0141492	px^-1
roots_per_100px_perimeter	normalised	3.0107	2.4134	1.8698	per 100 px
ring_attached_length_r30_px	network v2	179.294	147	134.016	px
ring_attached_length_r60_px	network v2	295.176	197	241.893	px
ring_length_w_r30_px	network v2	247.091	231.024	163.671	px
ring_length_w_r60_px	network v2	494.272	448.363	351.949	px
sholl_crossings_r10	network v2	4.9412	5	3.5964	count
sholl_crossings_r20	network v2	5.8235	6	4.0193	count
sholl_crossings_r30	network v2	6.5882	6	4.2435	count
loop_count	loops, report only	0.9412	0	2.2492	count
loop_edge_share	loops, report only	0.0684	0	0.1326	share
ring_width_mean_r30_px	width	6.8194	6.8038	0.8869	px

Per field
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measure	value	units
analysed_area_px2	993421	px^2
total_skeleton_length_px	17809	px
canalicular_length_density_per_px	0.0179269	px^-1
junction_count	353	count
junction_density_per_px2	0.00035534	px^-2
lacuna_count	21	count

lacunae_per_px2	2.114e-05	px^-2
skeleton_component_count	325	count
mean_component_length_px	54.7969	px
median_component_length_px	19	px
field_length_density_w_per_px	0.020067	px^-1
field_density_without_flagged_per_px	0.0169312	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1026	px
unattached_component_count	51	count
canalicular_length_density_attached_per_px	0.0179269	px^-1
field_length_density_with_unattached_per_px	0.0189597	px^-1
loop_count	30	count
loop_edge_share	0.1198	share
edge_count	1018	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.998	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	13594	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	582	37.2403	23.3143	1.5973	0.7798	0.8362	-1.3908	9.6323	718.978	True
2	556	37.5638	24.5507	1.5301	0.7569	0.7596	0.3365	28.2194	661.201	False
3	701	53.6823	22.5905	2.3763	0.9071	0.7242	1.3601	138.937	341.827	False
4	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
5	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
6	4692	101.01	74.3902	1.3578	0.6765	0.6687	0.5065	385.336	36.169	True
7	12231	202.474	100.48	2.0151	0.8682	0.7797	0.0478	526.667	38.6221	True
8	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
9	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
10	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
11	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
12	599	49.8148	18.0062	2.7665	0.9324	0.7799	1.3874	565.563	692.351	False
13	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
14	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
15	2065	77.2677	36.7006	2.1054	0.88	0.8765	-0.6191	783.319	730.515	False
16	2136	110.111	32.0301	3.4377	0.9568	0.7495	-0.4861	842.146	688.921	False
17	7245	127.692	88.342	1.4454	0.7221	0.8447	-1.2489	821.694	971.904	True
18	7049	221.75	78.7671	2.8153	0.9348	0.5533	-1.5418	863.529	297.474	False
19	2325	112.289	29.7925	3.769	0.9642	0.7992	1.5515	897.959	158.373	False
20	2028	91.0023	34.0428	2.6732	0.9274	0.8145	0.9697	910.357	523.35	False
21	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	1	89	171	17.4853	1	17.4853	6.933
2	5	213	313	378.635	26	14.5629	6.717
3	12	320	421	401.321	30	13.3774	7.496
4	8	218	510	837.512	41	20.4271	8.6503
5	8	343	755	809.838	50	16.1968	6.2878
6	9	347	687	1878.48	104	18.0623	7.0427
7	11	254	275	296.108	18	16.4504	7.1579
8	9	424	1083	1709.11	89	19.2035	7.0353
9	6	192	446	345.238	17	20.3081	6.6589
10	4	219	405	185.539	16	11.5962	6.8391
11	16	545	972	1116.52	59	18.924	7.842
12	5	141	237	166.255	8	20.7819	7.3126
13	7	348	751	1017.65	49	20.7683	6.0891
14	8	112	254	103.77	7	14.8242	6.8038
15	2	40	40	52.6274	4	13.1569	5.7532
16	5	73	124	46.2132	5	9.2426	6.3638
17	1	26	67	19.4853	1	19.4853	5.3649
18	5	330	688	680.801	31	21.9613	7.6111
19	3	121	254	120.284	6	20.0474	7.602
20	3	104	205	157.368	5	31.4735	5.7226
21	2	35	62	18.4853	1	18.4853	5.146

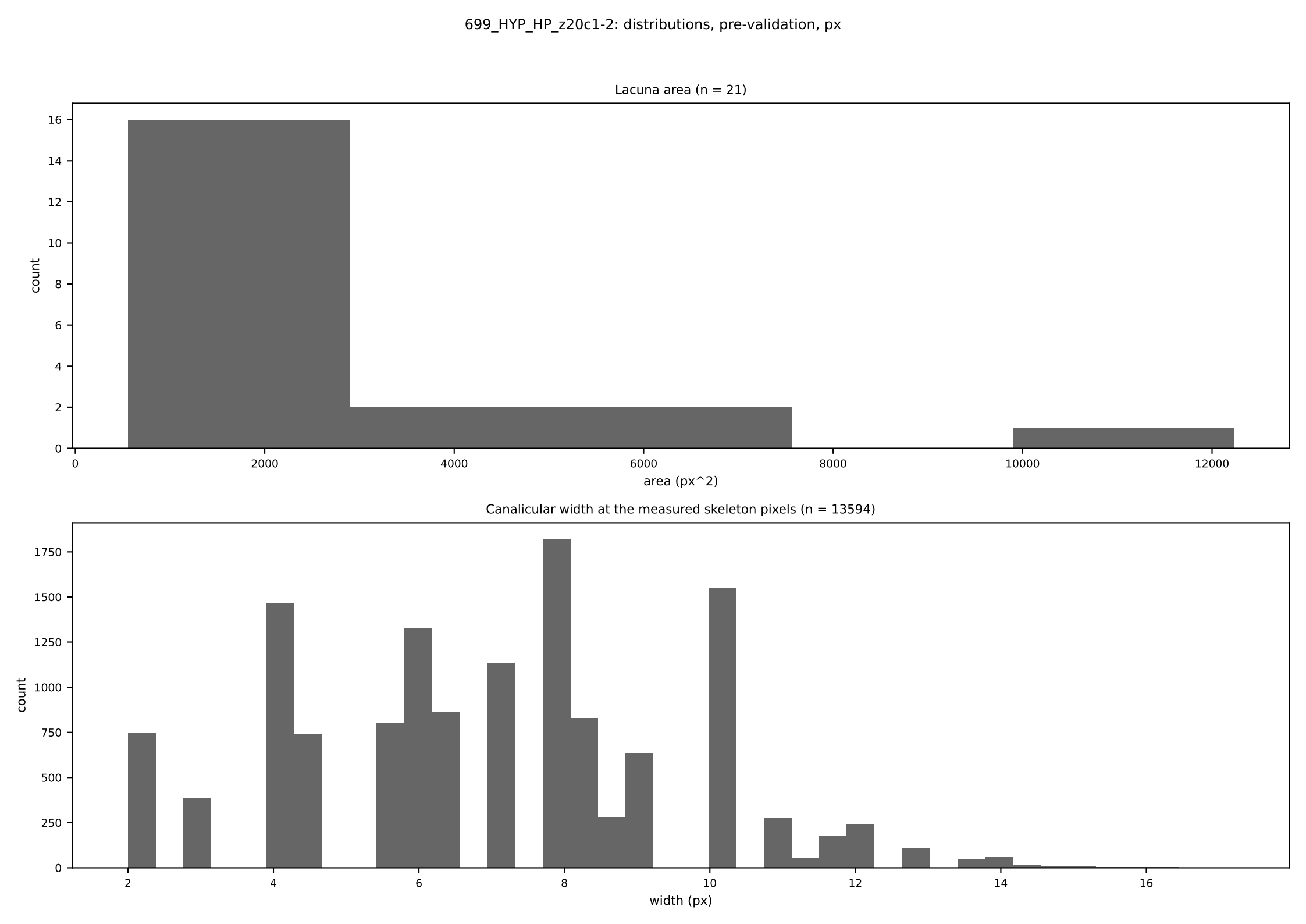
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	115.083	2908	6188	0.6007	0.566	0.0306052	0.0276341	0.8689
2	125.539	4509	10030	0.8517	0.729	0.0472389	0.0312064	3.9828
3	169.681	6393	14892	1	1	0.0500548	0.0282702	7.0721
4	185.581	6924	15979	1	1	0.0314847	0.0319169	4.3108
5	418.517	11182	24664	1	1	0.0306743	0.0306114	1.9115
6	534.86	8448	16501	0.7334	0.6731	0.0410748	0.0416338	1.6827
7	728.541	11456	21516	0.6459	0.6063	0.0221718	0.0127812	1.5099
8	221.179	7889	21117	1	1	0.0537457	0.0512857	4.0691
9	140.326	6761	16630	1	1	0.0283982	0.026819	4.2758
10	165.74	5442	13737	1	1	0.0402426	0.0294824	2.4134
11	316.149	10460	25779	1	1	0.0521032	0.0377051	5.0609
12	131.439	3919	7916	1	1	0.0359786	0.0299394	3.8041
13	132.267	6479	16157	1	1	0.053712	0.0464814	5.2923
14	363.747	9538	20842	1	1	0.0117425	0.0121869	2.1993
15	224.528	4228	7038	1	1	0.00946074	0.00568343	0.8908
16	285.22	8488	19536	1	1	0.00860038	0.00634726	1.753
17	652.181	10802	23498	0.7216	0.6773	0.00240696	0.00285131	0.1533
18	676.281	17978	35439	1	1	0.0183558	0.0194136	0.7393
19	308.551	9405	21041	1	1	0.0128655	0.0120717	0.9723
20	247.421	8982	20148	1	1	0.0115787	0.0101747	1.2125
21	163.681	5591	13507	0.7628	0.6662	0.00626006	0.00459021	1.2219

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	56	56	99.8406	189.539	3	2	2
2	147	197	231.024	337.451	4	6	4
3	295	348	355.25	468.576	8	8	6
4	205	428	244.238	573.345	5	7	8
5	216	332	386.433	843.11	9	9	12

6	306	474	386.35	769.671	8	5	8
7	248	269	287.037	308.764	7	7	3
8	353	769	474.262	1215.65	7	11	11
9	184	253	212.297	498.818	3	6	4
10	102	105	233.054	448.363	5	4	7
11	543	767	624.014	1112.56	15	13	12
12	139	164	151.912	260.338	5	4	4
13	298	637	387.25	841.541	6	12	13
14	112	182	120.498	274.238	4	2	5
15	40	40	42.5563	42.5563	1	1	0
16	73	91	82.7696	147.024	1	2	2
17	21	21	29.1421	74.7696	1	0	2
18	190	450	369.049	768.884	6	9	13
19	79	160	129.083	273.853	3	3	6
20	48	71	116.397	225.48	2	2	4
21	24	24	40.4558	70.8406	0	0	1


699_HYP_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 21)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
10000
12000
Canalicular width at the measured skeleton pixels (n = 13594)
count
width (px)
0
250
500
750
1000
1250
1500
1750
2
4
6
8
10
12
14
16